



ABES Engineering College, Ghaziabad
(An Autonomous Institute)

B.Tech. First Year, Semester-I Common to all branches)

Course Code	Course Name	L	T	P	C
	Basics of Electronics Engineering	3	1	0	4

Course Objective

To introduce first-year engineering students to the fundamental concepts of electronics, including electronic components, circuits, and latest emerging trends, laying the groundwork for advanced studies and practical applications in various engineering fields.

Course Outcomes (CO)	At the end of this course, student will be able to:	Bloom's Knowledge Level
CO1	Understand the working principle and characteristics of diodes and their various applications.	K3
CO2	Understand the working principle and characteristics of Transistors (BJT, FET & CMOS) & their applications.	K2
CO3	Understand the concepts of number systems, logic gates, binary arithmetic, Boolean algebra and Karnaugh maps for digital circuit design.	K3
CO4	Apply Op-amp & IC-555 in various practical applications.	K3
CO5	Understand the fundamentals & recent trends in Communication Engineering.	K2

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Program Outcomes → Course Outcomes ↓	CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)												Program Specific Outcomes PSO			
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	1	2	1	2	3		1		1				
CO2	2	2	2	1	2	1	2	3		1		1				
CO3	3	2	2	1	2	1	2	3		1		1				
CO4	2	2	2	1	2	1	2	3		1		1				

CO5	2	2	2	1	2	1	2	3		1		1				
Syllabus																
Unit-1	Diode & its Applications															10
PN Junction Diode: Working Principle, V-I Characteristics, Rectifiers - Half Wave & Full Wave, PIV, Ripple Factor and Efficiency, Clippers & Clampers																
Special Purpose Diodes: Construction, Characteristics and Applications - Tunnel Diode, Schottky Diode, Varactor Diode, Photodiode, Zener Diode																
Unit-2	Transistor and its Applications															12
BJT: Construction & Working Principle of Bipolar Junction Transistor, Common Base, Common Emitter and Common Collector Configurations and their characteristics, BJT as a switch																
FET: Comparison of BJT & FET; Construction, Working Principle and Characteristics of JFET & MOSFET (Enhancement and Depletion), Concept of pinch-off, FET as a voltage variable resistor and a constant current source																
CMOS: Construction, Working Principle and Characteristics of CMOS, NOT, NAND & NOR implementation using CMOS																
Unit-3	Digital Electronics															10
Number Systems, Conversions, Codes (Weighted and Non-weighted codes, BCD & Gray Code), Boolean Algebra, Complements, Binary Arithmetic, Logic Gates, Boolean Expressions (SOP & POS forms), Minimization using K-Map (up to 4 variables), don't care condition and realization using logic gates																
Sampling & Quantization in Digital Signal Processing																
Fundamentals and Applications of Quantum Computing and Edge Computing																
Unit-4	Basics of Integrated Circuits															
Introduction: Evolution from discrete components to IC, Moore's Law and Scaling, Monolithic v/s Hybrid IC, Levels of Integration (SSI, MSI, LSI, VLSI and ULSI)																

Overview of IC fabrication techniques: Wafer preparation, Oxidation, Photolithography, Etching, Doping (Diffusion and Ion implantation), Deposition, Metallization and Packaging		9
Op-Amp: Pin Configuration, Ideal Characteristics, Applications as Inverting Amplifiers, Non-Inverting Amplifiers		
555 Timer: Pin Configuration, Astable and Monostable operations, Applications as Waveform Generators and Delay Circuit		
Unit-5	Communication Technologies	7
Introduction: Elements of Communication System, Modulation, Need of modulation		
Cellular Networks: An overview, Different generations in cellular communication systems, Introduction to 5G and 6G		
Smart Technology: Introduction to IoT Architecture, Sensors and Transducers, Wireless Sensor Network		
Total No. of Lectures		48
Text Books		
1. Robert L. Boylestand / Louis Nashelsky “Electronic Devices and Circuit Theory”, Pearson Education		
2. M. Morris Mano, Michael D. Cileti, “Digital Design”, 6th Edition, Pearson Education.		
3. George Kennedy and Bernard Davis, “Electronic Communication Systems”, Tata McGraw Hill, Fourth Edition		
Reference Books		
1. Jacob Millman, C.C. Halkias, Satyabrata Jit, “Electronic Devices and Circuits”, McGraw Hill.□		
2. A. Anand Kumar, “Fundamental of Digital Circuits,” PHI 4th edition, 2018.		
3. D. P. Kothari and I. J. Nagrath, “Basic Electronics” McGraw Hill, 2nd Edition		
4. S Salivahanan and S Arivazhagan “Digital Electronics”, Vikas Publications		